

MARY T1 Expressed Affect: Artifact Ledger

BrainVI

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Scope

T1 is displayed affect: face and body signals present in the stimulus. Induced affect, physiology, BOLD, CASE, and viewer-state targets are out of scope.

Established face transfer

The current reported transfer cells and protocol are in `EXPRESSED-AFFECT-gold-transfer-results.md`. Pearson correlation is primary and CCC is secondary.

The locked AFEW-VA identity/group-CV result is recorded in `results/expressed_t1/expressed_id_t1.json`. Across five group folds, the commercial routed tier reaches mean Pearson $r = .643/.594$ for valence/arousal, with no output collapse and 10,000-draw matched nulls per fold. This is an in-domain identity/group robustness result, not LODO.

MPIIEmo research gate

The acquisition, license, ceiling, and actor-ownership protocol are in `T1-EXPRESSED-DATASET-LEDGER.md`. The recovered 448-trace ridge floor is negative in both disjoint evaluations: pair-held-out valence/arousal Pearson $r = -.068/-.112$, and scenario-held-out $r = -.357/-.311$. Matched circular-null p values are (.9997, .4969) and (.9356, .1192); neither split collapsed. This is a diagnostic failure floor, not a promoted training result; the full JSON remains authoritative.

Pillar 3

The body adapter is `experiments/scaling-sota-head/freemocap_kinematics.py`. It reports kinematics and quality diagnostics. The complete HEROES categorical smoke benchmark has 256 reconciled native-resolution outputs. The frozen ridge floor (temporal mean+SD pooling, source-only standardization) gives actor-held-out balanced accuracy/macro-F1 = .441/.442, scenario = .367/.365, and camera = .313/.312. The matched 10,000-draw within-group fixed-prediction label null gives $p = .0001$ for actor and scenario and $p = 1.0$ for camera. This is not a continuous BoLD result; the authoritative JSON is `results/heroes_t1/heroes_body_t1.json` and the durable archive is `s3://Neuropeer/oracle/research/scaling_sota_head/heroes_t1/freemocap_20260719/`.

The open-license MEED left-view benchmark is a separate categorical body result: 1,386 recordings from 22 actors and seven classes. Actor-held-out normalized 2-D pose/velocity ridge gives balanced accuracy/macro-F1 .539/.514 across three seeds. Its authoritative JSON is `results/ex`

pressed_t1/meed_body_t1_20260719.json (SHA-256 4a31422b5f07c1154140af80e926dc971f7b1ac84700c1ff4400133cfc626673). A fixed higher-order ablation adding acceleration, jerk, and expansiveness falls to .533/.514, so the added dynamics are retained as negative evidence under meed_body_t1_higher_order_20260719.json (SHA-256 0f58ddd168676041270fe4e14de836a62df81009f85eeb5c46b0cc3b61e433c9).

The Hanyang Emotional body motion control corpus adds 4,060 five-second, 19-joint 3-D trials from 29 participants and seven instructed emotions. The participant-held-out ridge floor reaches balanced accuracy/macro-F1 .447/.443 on intended labels and .468/.476 on observer-majority labels. The 13-rater mean per-motion majority agreement is .635; both model targets use 10,000 matched within-participant permutations. This is elicited, research-only mocap control evidence, not in-the-wild BoLD transfer. The authoritative artifact is results/expressed_t1/emotional_body_motion_t1_20260719.json (SHA-256 a8bc6f899ab1d53aca04517e78ef07d0206a4ef25547c98fcb2f46e15364e7a1).

The locked cross-representation stress test uses 18 common scalar kinematic features and the same seven labels. Hanyang→MEED macro-F1 is .083 and MEED→Hanyang is .046, below the seven-class reference .143; both predictions collapse to three labels. This is negative transfer evidence, traced to results/expressed_t1/body_motion_transfer_t1_20260719.json (SHA-256 886664caf8a2cfb21aeb75c974fc6ac88a867fc0b414b73ae11de3fcae6c0bf7).

The verified MPIIEmo continuous-body floor is separate from these categorical benchmarks. A normalized COCO17 2-D pose proxy with portrait identity gating reaches pair-held-out Pearson $r = -.038/.143$ and scenario-held-out $r = -.026/.085$ for valence/arousal. Matched circular-null p values are (.3108,.0028) and (.6793,.0005). The run produced 295 valid traces from 448 actor-owned videos; 153 unmatched videos remain in the manifests. This is research-only pose-proxy evidence, not a FreeMoCap or commercial full-coverage claim. The authoritative artifact is results/expressed_t1/mpiiemo_body_t1_20260719.json (SHA-256 ee8d884bfbb2798eef99ff496ff6c4afab9342b327d7fc9ed6d009e0f98825ab). Each cell includes 5,000 whole-group bootstrap draws in addition to the 10,000 matched nulls.

After both streams were independently evaluated, a matched face/body ridge ablation used the same 542 paired rows. Pair-held-out valence/arousal Pearson $r = .020/.129$ and scenario-held-out $r = -.033/.130$; matched-null p values are (.828,.240) and (.769,.0255). The scenario arousal cell does not replicate in the pair split, so fusion is retained as limited negative evidence and is not promoted. Its authoritative JSON is results/expressed_t1/mpiiemo_fusion_t1_20260719.json.

Temporal head comparison

The complete three-seed temporal comparison is traced in T1-TEMPORAL-HEAD-RESULTS.md. AFEW-VA→VEATIC GRU heads remain modestly positive (best mean valence $r = .219$, arousal $r = .223$), but VEATIC→AFEW-VA is negative on valence for every head (best mean $r = -.096$) and near-null on arousal. TCN+LSTM does not improve transfer robustness; the representation/domain shift remains the limiting failure.

Video-stream modality ablation

On the AFEW-VA→VEATIC transfer, source-only video-stream ablation gives mean Pearson valence/arousal = .121/.221 for SlowFast-only, .346/.316 for VL-only, and .351/.342 for fused video (three seeds each). The 10,000-draw target-clip circular null is recorded per seed in results/expr

essed_t1/modality_ablation_t1.json; audio and text are not ablated because they are absent from the AFEW source pool.

Artifact traceability

Every future headline number must cite its result JSON, prediction array, manifest, checksum, and evaluation command. The microexpression disposition is recorded in T1-MICROEXPRESSION-GATE.md: acquisition and rights are pending, with no model claim. The deception disposition is recorded in T1-DECEPTION-CUE-GATE.md. The T1-only runtime/package boundary is frozen in experiments/scaling-sota-head/production/t1_expressed_manifest.json; its body branch is research-only and T2/T3 are excluded.

Package readiness correction

The available research transfer checkpoints are preserved in research_transfer_checkpoint_manifest_t1_20260719.json (12 files, SHA-256 f35c6cad863afb2b6a0aa3cd58ccba51ba35e726969e2642342bd43b4db7e44d) and copied to the expressed-affect B2 checkpoint prefix. They are branch-level research transfer artifacts, not the combined routed inference artifact. The combined six-seed commercial-code package is separately inventoried under the legacy routed_expressed_v a/packaged/ prefix. The commercial package is therefore **not frozen for distribution**: AFEW-VA/VEATIC training rights remain unresolved. T1-DECEPTION-CUE-GATE.md: production lie detection is rejected at the current evidence gate, with no trained result. The official Michigan archive is locally intact (121 clips, 61 deceptive/60 truthful), but its files expose no clip-to-speaker mapping for the 56 speakers reported in the paper and do not clear redistribution. Its acquisition audit is results/expressed_t1/reallife_deception_acquisition_audit_20260719.json (SHA-256 5566445343c2569849171691ea7a45e970522ddd34f3683b03fdc5b8700a09a6); no model or calibration claim is made. The HEROES body numbers above are traced to results/heroes_t1/heroes_body_t1.json (SHA-256 92bc01b7a482292911093a0548dd291495f4563bee46a108e0ba1d009817a1c2) and the 256-entry checksum manifest results/heroes_t1/artifact_checksums.sha256. The identity result hash is 867902af53f1e1a18a90a9532671590650806fca3e213ae69f9e3f628e8c6351.